```
let grid = [];
for (let i = 0; i < 5; i++) {
  grid[i] = matrix.slice(i*5, i*5+5);
  }
return grid;
}

function findPos(matrix, ch) {
  for (let r = 0; r < 5; r++) {
  for (let c = 0; c < 5; c++) {
    if (matrix[r][c] === ch)
      return {r, c};
    }
  }
return null;
}

function processText(mode) {
let key = document.getElementById("globalKey")
  value;

let inputId = mode === "encrypt" ? "plaintext"
              "ciphertext-input"

let outputId = mode === "encrypt" ?
              "ciphertext-result",
              "plaintext-result";

let originalText = document.getElementById(inputId).value;
```



```
if (!key) {
  alert("configuration error");
  document.getElementById("globalKey").focus();
  return;
}

if (!originalText) return;

let matrix = generateKeyMatrix(key);

let preparedText = mode === "encrypt" ? formatPlainText
(originalText) : originalText.toUpperCase()
replace(/J/g, "I");

let result = "";

for (let i=0; i< preparedText.length; i+=2)
{
  let char1 = preparedText[i];
  let char2 = (i+1 < preparedText.length)
              preparedText[i+1] : 'X';

let posA = findPos(matrix, char1);

let posB = findPos(matrix, char2);

if (!posA || !posB) continue;
```


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```
let r1 = pos A.r, c1 = pos A.c;
let r2 = pos B.r, c2 = pos B.c;

if (r1 == r2) {
    let shift = mode == 'encrypt' ? 1 : 4;
result stream += matrix [r1] [(c1 + shift) % 5];

result stream += matrix [r2] [(c2 + shift) % 5];
}
    else if (c1 == c2) {
    let shift = mode == 'encrypt' ? 1 : 4;
result stream += matrix [(r1 + shift) % 5] [c2];
}
else {
    result stream += matrix [r1] [c2]
    result stream += matrix [r2] [c1];
}
}

let final output = result stream;
if (mode == 'Encrypt') {
document get Element By Id (output.Id)
    value = final output;
} else {
document get element by Id (output.Id).innerHTML
    Text = final output;
```

```
document.get element By Id (output.Id).Class
    list.add ("text-white");
}
}

function encrypt () { process Text ("encrypt");
}

function decrypt () { process Text ("decrypt");
}
```